

CHEMISTRY | UNIT 02 | LESSON 05

Periodic Trends and Chemical Properties

Notebook Evidence - Independent Application

Student name:		Date:	
MLA standard:	MLA.CHEM.ATO.03	Submission:	Notebook Evidence - Independent Application

Directions

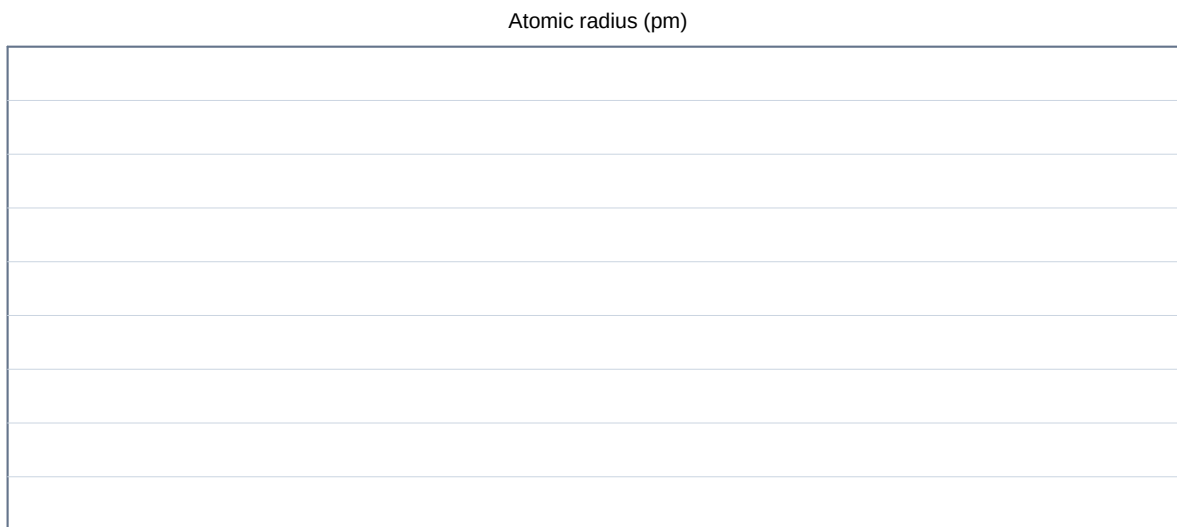
Use the Period 2 data below. Complete the organizer, plot atomic radius, and answer every analysis question. Submit this PDF in U2 L5 Notebook Evidence Submission.

Part A - Period 2 evidence

Element	Atomic #	Valence e-	Atomic radius (pm)	1st ionization energy (kJ/mol)	Electronegativity
Li	3	1	152	520	0.98
Be	4	2	112	900	1.57
B	5	3	85	801	2.04
C	6	4	77	1086	2.55
N	7	5	75	1402	3.04
O	8	6	73	1314	3.44
F	9	7	72	1681	3.98

Part B - Graph atomic radius

Plot atomic number on the horizontal axis and atomic radius on the vertical axis. Connect the points and give the graph a descriptive title.



Part C - Analyze the patterns

1. Describe the overall atomic-radius trend from lithium to fluorine and identify the evidence.

2. Describe the overall ionization-energy trend. Name the Be-to-B or N-to-O exception shown by the data.

3. Explain both trends using nuclear charge, shielding, and attraction between the nucleus and valence electrons.

4. Which element is most likely to attract shared electrons strongly? Support the answer with electronegativity evidence.
